

Extracting knowledge from textbooks: a full research landscape

Automatically extracting structured concept graphs from educational texts is now feasible through a convergence of cognitive learning theory, NLP extraction pipelines, and LLM-based methods — but the critical unsolved problem remains implicit concept detection. The field spans foundational cognitive theories (Ausubel, Novak, Bloom) that define *what* a concept hierarchy should look like, a rich set of explicit extraction tools (NER, keyphrase extraction, term recognition), a growing body of work on implicit/prerequisite concept discovery, and increasingly powerful LLM-based pipelines — each with distinct tradeoffs between correctness, explainability, and coverage. For a mind map application prioritizing verifiable correctness, the most promising architecture combines deterministic extraction methods with LLM reasoning under constrained decoding and symbolic verification, grounding every extracted concept and relation in traceable source spans.

1. Cognitive and representational foundations that define the target

Any concept extraction system ultimately reconstructs structures that mirror how knowledge is organized in human cognition. The theoretical foundations are not merely academic — they define what a "correct" extraction looks like.

Ausubel's subsumption theory (Ausubel, 1962, 1968) posits that cognitive structure is organized hierarchically, with inclusive concept clusters at the top under which less inclusive sub-concepts are *subsumed*. Four mechanisms govern learning: *derivative subsumption* (new facts fit under existing concepts without altering them), *correlative subsumption* (new information extends existing concepts), (Mlrg) *superordinate learning* (a new general concept emerges to encompass known specifics), and *combinatorial learning* (new concepts relate to general background without direct subsumption). (Mlrg) Ausubel's core claim — "the most important single factor influencing learning is what the learner already knows" — directly motivates prerequisite detection algorithms. (iMindQ)

Novak's concept map theory (Novak & Gowin, 1984; Novak & Cañas, 2006) operationalized Ausubel's ideas into a formal graphical representation: nodes represent concepts ("perceived regularities in events or objects, designated by a label"), labeled directed edges represent propositions linking two concepts with a relationship phrase, (Northwestern) and maps are organized hierarchically from general to specific (ResearchGate) with cross-links showing integrative reconciliation across branches. (Northwestern) This structure — nodes, labeled edges, hierarchy, cross-links — is the natural target representation for automated concept map generation, with roughly 50 published systems attempting it between 2001 and 2020 (Nugumanova et al., 2021). (arXiv)

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Frame semantics (Fillmore, 1982; Baker et al., 1998) provides a linguistic foundation. A semantic frame is a schematized situation (e.g., the Commercial_Transaction frame with buyer, seller, goods, money), evoked by lexical units and populated by frame elements. **FrameNet** catalogs over **1,200 frames and 13,000 lexical units** (ResearchGate) with 200,000+ annotated sentences. (Springer) For concept extraction, frames provide structured templates that organize domain knowledge beyond flat entity lists. The related work by **Schank** (1972, 1977) on conceptual dependency theory — with 11 primitive acts (ATRANS, PTRANS, MTRANS, etc.) (Slideshare) and stereotyped scripts (the restaurant script) (U-picardie) — demonstrated that knowledge can be decomposed into language-independent primitives, (Wikipedia) a precursor to modern ontology engineering.

Ontology learning from text follows the six-layer "ontology learning layer cake" (Cimiano, 2006; Buitelaar et al., 2005): terms → synonyms → concepts → concept hierarchies/taxonomies → non-taxonomic relations → axioms/rules. (PubMed Central) (ResearchGate) Each layer demands different NLP techniques: term extraction uses C-value and TF-IDF, (ResearchGate) taxonomy building uses Hearst patterns and distributional methods, and relation extraction uses dependency parsing and association rule mining. (Oxford Academic) The Text2Onto system (Cimiano & Völker, 2005) (Springer) and its predecessors (Maedche & Staab, 2001) remain foundational references. (PubMed Central)

For **knowledge representation**, three formalisms are most relevant. Quillian's semantic networks (1967) introduced labeled graph structures (ScienceDirect) with property inheritance and spreading activation (ResearchGate) — the direct ancestor of modern knowledge graphs. Sowa's conceptual graphs (1984) combine semantic networks with formal logic, (Jfsowa) mapping concept nodes [Type: Referent] and relation nodes through canonical formation rules to first-order logic (standardized as ISO/IEC 24707). Description logics (Baader et al., 2003) provide the formal underpinning for OWL ontologies, with TBox/ABox architecture enabling automated reasoning — subsumption checking, classification, and consistency verification of extracted knowledge.

Two educational theories complete the picture. **Bloom's revised taxonomy** (Anderson & Krathwohl, 2001) creates a two-dimensional matrix of cognitive processes (Remember → Understand → Apply → Analyze → Evaluate → Create) crossed with knowledge types (Factual, Conceptual, Procedural, Metacognitive). (PubMed Central) Higher Bloom levels correspond to more abstract concept understanding, and automated Bloom-level classification of text segments has achieved **94% accuracy** with SVM plus synonym augmentation (2024-2025 studies). (arXiv) **Threshold concepts theory** (Meyer & Land, 2003) identifies concepts that are transformative, irreversible, integrative, bounded, and troublesome (Ucalgary) — concepts like "opportunity cost" in economics or "recursion" in computer science that permanently restructure a learner's understanding. No fully automated system for threshold concept identification yet exists, representing a significant research gap.

2. Explicit concept extraction: the established toolkit

Explicit concept extraction identifies concepts that are directly stated in text. The methods fall into four families with distinct explainability profiles.

Named Entity Recognition for scientific text has matured around the SciERC benchmark (Luan et al., 2018): 500 annotated AI/ML/NLP abstracts ([University of Washington](#)) with six entity types (Task, Method, Metric, Material, Other-Scientific-Term, Generic) and seven relation types. The best supervised NER models achieve **F1 of ~70–83%** on SciERC, with PL-Marker (Ye et al., 2022) and HGERE (2024) pushing the state of the art. **SciBERT** (Beltagy et al., 2019), pretrained on 1.14M Semantic Scholar papers, ([CEUR-WS.org](#)) and **SciSpaCy** (Neumann et al., 2019) provide domain-adapted models with entity linking to UMLS, MeSH, and similar resources. The central challenge for educational text is that "concepts" (gradient descent, electromagnetic induction) are not traditional named entities — they have fuzzy boundaries, are often multi-word, and are frequently nested.

Keyphrase extraction offers four well-characterized unsupervised methods. **TextRank** (Mihalcea & Tarau, 2004) builds a co-occurrence graph of nouns/adjectives and applies PageRank, achieving F1@10 of ~10–15% on exact match. **RAKE** (Rose et al., 2010) splits text at stopwords, scores candidates by word degree/frequency ratio, and runs extremely fast (~2,000 documents in 2 seconds). **YAKE** (Campos et al., 2020) combines five statistical features (casing, position, frequency, context relatedness, sentence distribution) and **outperforms TF-IDF, RAKE, TextRank, and even supervised KEA on 100% of 20 tested datasets** — the strongest unsupervised statistical method. **KeyBERT** (Grootendorst, 2020) generates SBERT embeddings for candidates and the full document, ranking by cosine similarity with Maximal Marginal Relevance diversification; it achieves the highest accuracy on long texts but is significantly slower. All four are fully deterministic and explainable (TextRank graphs, YAKE feature scores, KeyBERT similarity scores are all inspectable).

Automatic Term Recognition addresses a different task: identifying domain-specific terminology based on *termhood* (is this a technical term?) rather than *document representativeness*. The **C-value/NC-value method** (Frantzi et al., 2000) remains foundational — it applies linguistic filters (Noun+Noun, Adj+Noun patterns), penalizes terms that appear primarily as substrings of longer terms, and weights by contextual co-occurrence. With 1,500+ citations, it is the standard baseline for multi-word term extraction in domain-specific corpora.

For **STEM textbooks specifically**, the most notable systems include **FACE** (Chau et al., 2021), a supervised method combining TF-IDF, domain relevance, structural features (section position, heading presence), and Wikipedia-based features that outperforms TextRank and RAKE on computer science textbook data, and **WERECE** (2023), an unsupervised method using manifold learning to refine word embeddings for educational concept extraction, achieving **precision 85.9%, recall 87.0%, F1 86.4%** on educational datasets — the current state of the art for

unsupervised educational concept extraction.

GLiNER (Zaratiana et al., 2024, NAACL) enables zero-shot NER by framing entity recognition as a matching problem: a DeBERTa-v3 encoder processes concatenated entity type labels and input text, computing matching scores between entity type embeddings and span representations. GLiNER-Large achieves **zero-shot F1 of ~54-61% on CrossNER** (53% on AI domain, 58% on Science) using only ~304M parameters — comparable to UniNER-13B at 140× smaller size. The GLiNER-bi-Encoder variant (Stepanov et al., 2026) scales to 1,000+ entity types with **61.5% Micro-F1** and 130× throughput improvement. For a mind map application, GLiNER allows specifying custom entity types ("scientific concept," "mathematical term," "biological process") without any training data — ideal for rapid prototyping across domains.

Ontology-based extraction constrains and validates extraction using pre-existing knowledge structures. Dictionary/gazetteer matching (e.g., MetaMap mapping biomedical text to UMLS's 4M+ concepts), pattern-based extraction guided by ontology structure, and embedding-based fuzzy matching against ontology labels all provide higher precision through known-vocabulary constraints. The tradeoff is coverage: novel or emerging concepts absent from the ontology will be missed.

3. Implicit concept extraction: the critical frontier

Implicit concept extraction — detecting concepts that are structurally essential but not explicitly stated, prerequisite relationships that govern learning order, and abstract organizing principles — is the hardest and most impactful problem for educational mind maps.

Detecting structurally essential "unifying" concepts

A concept like "limit" in calculus is mentioned explicitly in Chapter 2 but implicitly governs derivatives (Chapter 3), integrals (Chapter 4), and series (Chapter 5). Detecting such concepts requires graph-based centrality analysis on extracted concept networks. **Betweenness centrality** (Freeman) measures how often a concept lies on shortest paths between other concepts — high betweenness with moderate mention frequency signals a bridging/unifying concept. **PageRank** assigns importance based on incoming links from other important concepts. **Eigenvector centrality** captures importance as a function of neighbors' importance. The practical pipeline: extract concept terms using NER/keyphrase methods, build a concept co-occurrence or dependency graph, compute centrality metrics, and identify concepts with **high betweenness but moderate or low explicit frequency** — these are the implicit governing concepts. Threshold concepts (Meyer & Land, 2003) provide the theoretical frame: [Springer](#) computationally, threshold concepts should exhibit high betweenness centrality, high cross-section reach, and position as bridges between otherwise disconnected concept clusters.

Prerequisite detection: the most developed implicit extraction area

Prerequisite detection on the most developed implicit extraction area. Automatically identifying that "limits" must precede "derivatives" has attracted substantial research since 2011.

RefD (Reference Distance) (Liang et al., 2015, EMNLP) is the foundational unsupervised method. (ACM Digital Library) It measures asymmetric reference patterns between Wikipedia concept articles: (Academia.edu) if most concepts related to B link to A, but few related to A link to B, then A is likely a prerequisite of B. (ACM Computing Surveys) RefD outperformed supervised methods across 7 domains on 1,336 concept pairs — a remarkable result for a single-metric unsupervised approach. **Pan et al. (2017, ACL)** extended RefD with Video Reference Distance and Sentence Reference Distance from MOOC subtitles, (Tsinghua) (1Library) significantly outperforming the original on Coursera ML, DSA, and Calculus datasets. (ACL Anthology) (Tsinghua) **PREREQ** (Roy et al., 2019, AAAI) uses Pairwise-Link LDA for latent topic representations followed by a Siamese Network classifier, effective with very limited labeled data. (Academia.edu +2) **iPRL** (Lu et al., 2019, AAAI) iteratively combines learning-based and recovery-based models, (ACM Digital Library) improving F1 by **12.85% and 18.39%** on ML and DSA datasets over baselines. (AAAI) Most recently, **TCPL** (2023) performs continual pre-training with domain-adaptive BERT plus R-GCN on heterogeneous concept-resource graphs, (MDPI) and **GKROM** (2025, AAAI) uses multi-objective optimization for global knowledge relation networks. (AAAI)

Key benchmark datasets include the University Course Dataset (Liang et al., 2017: 1,336 CS/Math concept pairs from 11 U.S. universities), (Semantic Scholar) (ACM Digital Library) **AL-CPL** (Liang et al., 2018: 6,529 KC pairs across Data Mining, Geometry, Physics, Precalculus), (ACM Computing Surveys) and **MOOCCube** (Yu et al., 2020: 114,563 concepts from 706 courses). (ACL Anthology) The field received a comprehensive survey in *ACM Computing Surveys* (2025).

Cross-section reasoning and concept propagation

Tracking implicit concept presence across chapters requires discourse-level NLP. State-of-the-art within-document coreference achieves **~80% CoNLL F1** on OntoNotes (Kirstain et al., 2021), but cross-document/book-scale coreference remains challenging (~36% F1 for full books per Guo et al., 2023). **BookCoref** (2025) provides a pipeline for book-scale coreference using entity linking for cluster initialization. For textbooks, the practical approach combines explicit mention tracking with embedding-based similarity: a concept is implicitly present when its closely related terms appear, even without direct mention. Building a concept × section presence matrix (explicit + implicit) reveals concepts with high cross-section reach but low explicit frequency — structurally essential implicit concepts.

Latent concept discovery through topic modeling

LDA (Blei et al., 2003) discovers latent topics as distributions over words, (BERTopic) (GeeksforGeeks) applicable to identifying concepts that emerge from word co-occurrence patterns rather than explicit mention. **Hierarchical LDA** (Blei et al., 2003/2004, NeurIPS) uses a nested Chinese Restaurant Process to discover topic hierarchies of arbitrary depth (Columbia University)

Restaurant Process to discover topic hierarchies of arbitrary depth (Columbia University) — higher-level topics are more general/abstract, naturally discovering organizing principles not explicitly named. **BERTopic** (Grootendorst, 2022) replaces LDA's bag-of-words with SBERT embeddings → UMAP → HDBSCAN clustering → class-based TF-IDF, producing more coherent topics. (Chamomile) Najmani et al. (2023) showed BERTopic produces more coherent MOOC topics than LDA by c_v coherence. (Springer) For abstraction discovery, hierarchical topic models are particularly powerful: topics that appear at high levels of the hLDA tree correspond to implicit organizing concepts.

Knowledge graph completion for predicting unstated relationships

Once an initial concept graph is built, embedding-based KG completion can predict missing edges. **TransE** (Bordes et al., 2013) models relations as translations ($h + r \approx t$) but struggles with complex relations. **RotatE** (Sun et al., 2019, ICLR) models relations as rotations in complex space, achieving **MRR ~0.338 on FB15k-237 and ~0.476 on WN18RR**. Graph neural network methods like **R-GCN** (Schlichtkrull et al., 2018) and (arXiv) **GraIL** (Teru et al., 2020) enable inductive reasoning over unseen entities. (ScienceDirect) For educational KGs specifically, **MHAVGAE** (Zhang et al., 2022) applies multi-head attention variational graph auto-encoders to concept prerequisite prediction, (ACM Digital Library) and **Jia et al. (2021, NAACL)** use heterogeneous GNNs for prerequisite relation learning. (AAAI) (ACM Digital Library)

Rhetorical Structure Theory reveals discourse-encoded implicit knowledge

RST (Mann & Thompson, 1988) represents texts as hierarchical trees where Elementary Discourse Units are connected by rhetorical relations (Elaboration, Cause, Background, Condition). In textbooks, RST structure directly encodes implicit conceptual dependencies: Background relations signal prerequisite knowledge, Elaboration chains show progressive concept development, and Condition/Means relations encode procedural prerequisites. Modern RST parsers (Nguyen et al., 2021, "RST Parsing from Scratch") achieve state-of-the-art performance on RST-DT without handcrafted features.

Educational data mining closes the loop with learner evidence

When student performance data is available, (Wikipedia) **COMMAND** (Chen et al., 2016, EDM) jointly infers prerequisite graphs and student models using Structural Expectation Maximization over Bayesian networks. **Learning Factors Analysis** (Cen et al.) automatically discovers cognitive models from student data that consistently outperform human-created models. (Cmu) Probabilistic association rule mining on student performance data (Chen & Tian, 2015) reveals prerequisite structures through error propagation patterns: (ERIC) when students fail concept B but succeed on concept A, the asymmetry suggests A is prerequisite to B.

4. Relation extraction: connecting concepts into structured knowledge

Relation extraction: connecting concepts into structured knowledge

Beyond extracting isolated concepts, building a mind map requires identifying the *relations* between them. Each relation type demands different extraction methods.

Taxonomic (is-a) relations remain best served by **Hearst patterns** (Hearst, 1992): "NP_Y such as NP_X," "NP_X is a kind of NP_Y," and four other lexico-syntactic patterns. Crucially, Roller et al. (2018, ACL) showed that simple Hearst patterns with SVD smoothing **consistently outperform distributional methods** (WeedsPrec, invCL, SLQS) on all standard benchmarks (ACL Anthology) (BLESS, LEDS, EVALuation, WBLESS) — a striking vindication of deterministic, explainable methods. (aclanthology) For taxonomy induction from scratch, **TaxoGen** (Zhang et al., KDD 2018) performs unsupervised hierarchical clustering with adaptive term embeddings, (arXiv) and **TaxoRL** (Mao et al., ACL 2018) uses reinforcement learning for optimal edge addition.

Causal relation extraction has reached high performance on standard benchmarks: BERT models achieve **97.68% F1** on SemEval-2010 Task 8 (Khetan et al., 2020), (CEUR-WS.org) while inter-sentential causal extraction remains harder at ~66% F1 (Dasgupta et al., 2018; (ResearchGate) Verga et al.). Pattern-based methods ("X causes Y," "X leads to Y," "because of X, Y") are fully explainable and effective in educational text where causal language tends to be explicit.

Joint entity-relation extraction models handle both tasks simultaneously. **PL-Marker** (Ye et al., 2022, ACL) uses "levitated markers" with packed attention for span representation, achieving **4.1-4.3% strict F1 improvement** over previous state of the art on ACE04/05 (arXiv) and SOTA on SciERC — making it the best model for scientific text. **OneRel** (Shang et al., 2022, AAAI) casts extraction as 3D triple classification (arXiv) with 6.8× training speed improvement over TPLinker. (arXiv) **REBEL** (Huguet Cabot & Navigli, 2021, EMNLP Findings) uses BART-based seq2seq to generate linearized triplets, handling 200+ relation types simultaneously. (Babelscape srl) For educational text, **PL-Marker with SciBERT** is the strongest existing option due to its SciERC performance.

Open Information Extraction systems extract arbitrary relation triples without predefined schemas. The evolution from TextRunner (Banko et al., 2007) through **ReVerb** (Fader et al., 2011; (Microsoft) >30% extractions at precision ≥0.8) to **Stanford OpenIE** (Angeli et al., 2015; (ResearchGate) ~100 sentences/sec) has made OpenIE practical. (CoreNLP) Neural OpenIE (Cui et al., 2018; DocIE, 2021) achieves best F1 across domains. However, OpenIE produces open-vocabulary relations requiring significant post-processing for structured concept maps: entity normalization, relation type classification, and pedagogical relevance filtering.

For **cross-document relation extraction**, the CodRE dataset (Yao et al., 2021) provides the first human-annotated benchmark. Recent systems like **RAKG** (2025) achieve **95.81% accuracy** in document-level KG construction via RAG-enhanced cross-document linking, substantially outperforming KGGen (86.48%) and GraphRAG (89.71%). (arXiv)

5. LLM-based methods: power with hallucination risk

LLMs have transformed concept extraction but introduce hallucination risks that demand systematic mitigation.

Dagdelen et al. (2024, *Nature Communications*) demonstrated that fine-tuned GPT-3 and Llama-2 match or exceed traditional IE pipelines for scientific NER and relation extraction in materials chemistry, with outputs requiring very little human correction at **≥50 training examples** (Nature) — a landmark result with 485 citations. Zero-shot LLM NER using GPT-4 achieves ~55.6% F1 on clinical benchmarks, while supervised models still lead at 70–83% on SciERC. The critical advantage of LLMs is cross-document, cross-modal reasoning that traditional sentence-level extractors cannot perform.

Chain-of-thought prompting (Wei et al., 2022, NeurIPS) improves extraction quality through step-by-step reasoning. (arXiv) **CoT-ER** (2023) applies three-step reasoning — identify entities, map to concept hierarchy, extract contextual evidence — achieving competitive results on FewRel benchmarks. (arXiv) A critical finding from **Tam et al. (EMNLP 2024)**: constrained JSON decoding can *degrade* reasoning if answer fields precede reasoning fields. The fix is to always place (reasoning) and (evidence) fields *before* (answer) fields in output schemas. (Let's Data Science)

For **KG construction**, **GraphRAG** (Edge et al., 2024, Microsoft) uses community detection on LLM-extracted entities, (ResearchGate) (Medium) claiming **90% hallucination reduction** versus traditional RAG. (Medium) **Ontogenia** (Lippolis et al., 2025) uses metacognitive prompting for ontology generation with self-reflection. (arXiv) Feng et al. (2024) showed that "ontology-grounded extraction" — first generating a domain ontology, then using it as a directive prompt — improves both precision and recall. (arXiv)

Hallucination rates remain concerning. In clinical domains, LLMs show **50–82% hallucination rates** when elaborating on medical content (Omar et al., 2025, *Nature Communications Medicine*). (Nature) Even with RAG, RAGTruth (Niu et al., 2024, ACL) found LLMs produce unsupported or contradictory claims, with GPT-4-turbo achieving only **24% span-detection precision** for hallucination identification. The Vectara Hallucination Leaderboard (2025) reports that even the best models (Grok 4.20 Beta at 22%, Claude 4.5 Haiku at 25%) hallucinate on a substantial fraction of responses. (Artificial Analysis)

Five-layer defense against hallucination

The research points to a defense-in-depth strategy:

- **RAG grounding:** Self-RAG (Asai et al., 2024, ICLR top 1% oral) adaptively retrieves passages using reflection tokens, outperforming ChatGPT and retrieval-augmented Llama2-chat on factuality and citation accuracy. Modular RAG architectures achieve **attribution accuracy of 0.91**.
- **Constrained decoding:** Tools like **Outlines** (Willard & Louf, 2023), **Instructor** (Liu, 2022),

and **XGrammar** (Dong et al., 2024) compile JSON schemas into finite-state machines (arXiv) for O(1) valid token lookup, guaranteeing syntactic validity — though not semantic correctness. (Let's Data Science)

- **Self-consistency voting:** Wang et al. (2023, ICLR) sample multiple reasoning paths and take majority vote. CISC (ACL 2025) adds confidence-weighted voting.
- **Self-verification:** Gero et al. (2023, Microsoft Research) showed that asking LLMs to provide provenance for their own extractions and then verify them exploits the asymmetry between verification (easier) and generation (harder), consistently improving accuracy.
- **Neuro-symbolic verification:** Prenosil et al. (2025, *Nature Communications Medicine*) demonstrated that GPT-4 extraction followed by rule-based verification against a medical ontology achieves **100% accuracy** on ontology-covered parameters — versus 94.7% for GPT-4 alone. (PubMed Central) **VerifNER** (Kim et al., ACL 2024) improved NER F1 from 0.622 to 0.704 by verifying against UMLS.

Multi-agent verification provides additional reliability. **RECONCILE** (ACL 2024) runs round-table discussion among diverse LLMs (GPT-4, Claude, Bard) with confidence-weighted consensus, outperforming 9-way self-consistency. **BoN-MAV** (2025) shows that multiple small verifiers can improve even stronger generators through weak-to-strong generalization. (arXiv)

6. Educational technology applications and mind map generation

Automatic concept map generation has progressed through three generations. Rule-based systems like **Concept Map Miner** (Villalón & Calvo, 2008/2011) extract compound nouns via grammatical parse trees and identify relationships through a semantic layer. (Semantic Scholar) **TEXCOMON/OntoCmaps** (Zouaq & Nkambou, 2008/2009) uses deep semantic analysis with syntactic dependency patterns, building concept maps assessed via structural metrics (degree/betweenness/eigenvector centrality). (Academia.edu) **Falke et al. (2017, EMNLP/IJCNLP)** treat concept map generation as multi-document summarization: OpenIE extracts propositions, concept coreference resolution merges duplicates, and Integer Linear Programming finds optimal summary maps. (Semantic Scholar) (ACL Anthology) The most recent approaches use fine-tuned LLMs to extract structured propositions directly, (Sociedade Brasileira de Co...) achieving improving but still modest automated scores against expert-crafted gold standards.

Educational knowledge graphs provide large-scale infrastructure. **MOOCCube** (Yu et al., 2020, ACL) contains **706 courses, 114,563 concepts, and 8M+ behavioral records** (ACL Anthology) from XuetangX, with 8 entity types and 12 relation types including prerequisites. (PubMed Central) **MOOCCubeX** (2021) extends this to 4,216 courses, 637,572 concepts, and 296M behavioral records from 3.3M students. (ACM Digital Library) **KnowEdu** (Chen et al., 2018) constructs mathematics KGs through GPU-based neural sequence labeling plus probabilistic association

rule mining on assessment data. [Semantic Scholar](#) [IEEE Xplore](#) **CS-KG 2.0** (2025) describes 25M entities linked by 67M relationships from 15M computer science papers. [Nature](#)

Knowledge tracing reveals concept structure from student data. **BKT** (Corbett & Anderson, 1995) tracks binary knowledge states per skill [Nature](#) with four parameters. **DKT** (Piech et al., 2015, NeurIPS) uses LSTMs to model temporal dynamics, achieving **AUC of 0.86** on Assistments — a 25% gain over prior best. [Stanford University](#) Critically, DKT's conditional influence graphs between exercises reveal perfect clustering of hidden concepts, naturally producing concept dependency structures. [Stanford University](#) Graph-based extensions like **GKT** (Nakagawa et al., 2019) explicitly model prerequisite structures between knowledge components. [OpenReview](#)

Curriculum sequencing applies topological sorting to prerequisite DAGs, with optimization for multiple orderings. The KGCD framework models concept interdependencies as weighted directed edges, achieving **85% curriculum coherence** versus 60% in control groups. [Americaspj](#) Recent reinforcement learning approaches (CSEAL, RL-Tutor, DLELP) integrate knowledge tracing with adaptive path recommendation, [arXiv](#) and MWO (2025) uses a Memetic Walrus Optimizer with three-tier priority (concept coverage, difficulty progression, prerequisites) achieving **95.3% difficulty progression rate**. [arXiv](#)

7. Evaluation: measuring what matters for correctness

Evaluating concept extraction uses standard NER metrics (precision, recall, F1) [ScienceDirect](#) with strict match (identical spans) or lenient match (partial overlap). Gold standard creation typically requires 2+ domain expert annotators with adjudication; **pairwise F1** is the recommended inter-annotator agreement measure for span-based annotations because it avoids the ill-defined negative case count problem inherent in Cohen's κ for entity recognition (Hripcsak & Rothschild, 2005). [Prodigy](#)

The key benchmark datasets for educational concept extraction are:

- **SciERC** (Luan et al., 2018): 500 scientific abstracts, 6 entity types, 7 relation types, coreference. [CEUR-WS.org](#) SOTA NER F1 ~70–83%, relation F1 ~50.84%. [arXiv](#)
- **SciER** (2024, EMNLP): 24K entities, 12K relations from 106 full papers — much larger than SciERC. [arXiv](#)
- **AI-KG** (Dessi et al., 2020, ISWC): 820K entities, 14M triples from 333K AI publications. [Open University](#) [ACM Digital Library](#) Now superseded by CS-KG 2.0. [Open](#)
- **TutorialBank** (Fabbri et al., 2018, ACL): ~6,500 NLP resources [Alexander R. Fabbri](#) with prerequisite chain annotations.

- **MOOCCube/MOOCCubeX**: Concept and prerequisite annotations at scale. [PubMed Central](#)
[ACL Anthology](#)
- **AL-CPL**: 6,529 knowledge component pairs across 4 domains for prerequisite learning evaluation. [ACM Computing Surveys](#)

For concept map evaluation specifically, the **Novak-Gowin scoring scheme** (1984) awards points for valid propositions (1 pt), hierarchy levels (5 pts), branches (1 pt), cross-links (10 pts), and examples (1 pt) — heavily weighting hierarchy and integration. [University of Waterloo](#) The Waterloo Rubric standardizes this for automated assessment, with the **Kastor system** (Bhatia et al.) achieving **84–95% agreement** with human rubric scores. [Sacred Heart University](#) The key challenge remains concept granularity: annotators disagree on whether "natural language processing" and "natural language" are one concept or two, and whether domain-specific terms (e.g., "backpropagation") should be extracted at different specificity levels for different audiences.

A recommended architecture for correctness-first mind map extraction

Synthesizing across all seven research areas, the highest-reliability pipeline for extracting mind maps from textbooks would proceed in stages, each with verification.

Stage 1 — Explicit concept extraction. Run YAKE (deterministic, fast, best unsupervised statistical method) + C-value (for multi-word terms) + GLiNER (zero-shot, custom entity types) in parallel. Take the union, deduplicate by embedding similarity, and validate against domain ontologies where available.

Stage 2 — Latent concept discovery. Apply BERTopic for section-level topic modeling to discover concepts not captured by explicit extraction. Use hLDA for hierarchical topic organization to surface abstract organizing principles.

Stage 3 — Relation extraction. Apply Hearst patterns for taxonomic relations (fully deterministic, proven to outperform distributional methods per Roller et al., 2018). [arXiv](#) Use PL-Marker with SciBERT for joint entity-relation extraction on scientific text. Apply RefD against Wikipedia for prerequisite relations as a baseline.

Stage 4 — Implicit structure detection. Build the concept graph, compute betweenness centrality and PageRank to identify governing concepts. Track concept propagation across sections via embedding similarity. Apply RST parsing to identify discourse-encoded prerequisites (Background, Elaboration relations).

Stage 5 — LLM-augmented extraction with verification. Use GPT-4-class models with RAG grounding + structured output (Instructor/Outlines) + CoT prompting (reasoning before answers) to extract additional concepts and relationships. Verify every LLM extraction against

answers) to extract additional concepts and relationships. verify every LLM extraction against source spans (self-verification), cross-check against the deterministic pipeline output, and validate against domain ontology constraints. Apply self-consistency (3–5 samples, majority vote) for uncertain extractions.

Stage 6 — Graph completion and sequencing. Apply RotatE or R-GCN for link prediction to surface high-confidence unstated relationships. Topologically sort the prerequisite DAG for curriculum sequencing. Position high-centrality concepts as hub nodes in the mind map.

This architecture prioritizes **deterministic, explainable methods as the backbone** (YAKE, Hearst patterns, RefD, centrality metrics) while using LLMs as a supplementary layer with systematic verification — achieving the correctness and explainability the application demands without sacrificing the coverage advantages of neural methods.